

Course Overview

Welcome to Introduction to Machine Learning! This course will provide you with a comprehensive understanding of machine learning concepts, algorithms, and practical applications.

Course Structure

- 10 sessions of 4 hours each
- Combination of theory and practice
- Hands-on exercises using Python
- Interactive elements (polls, cloud numbers)

Learning Objectives

By the end of this course, you will:

- Understand fundamental ML concepts
- Be able to implement basic ML algorithms
- Know how to evaluate ML models
- Have practical experience with real-world datasets

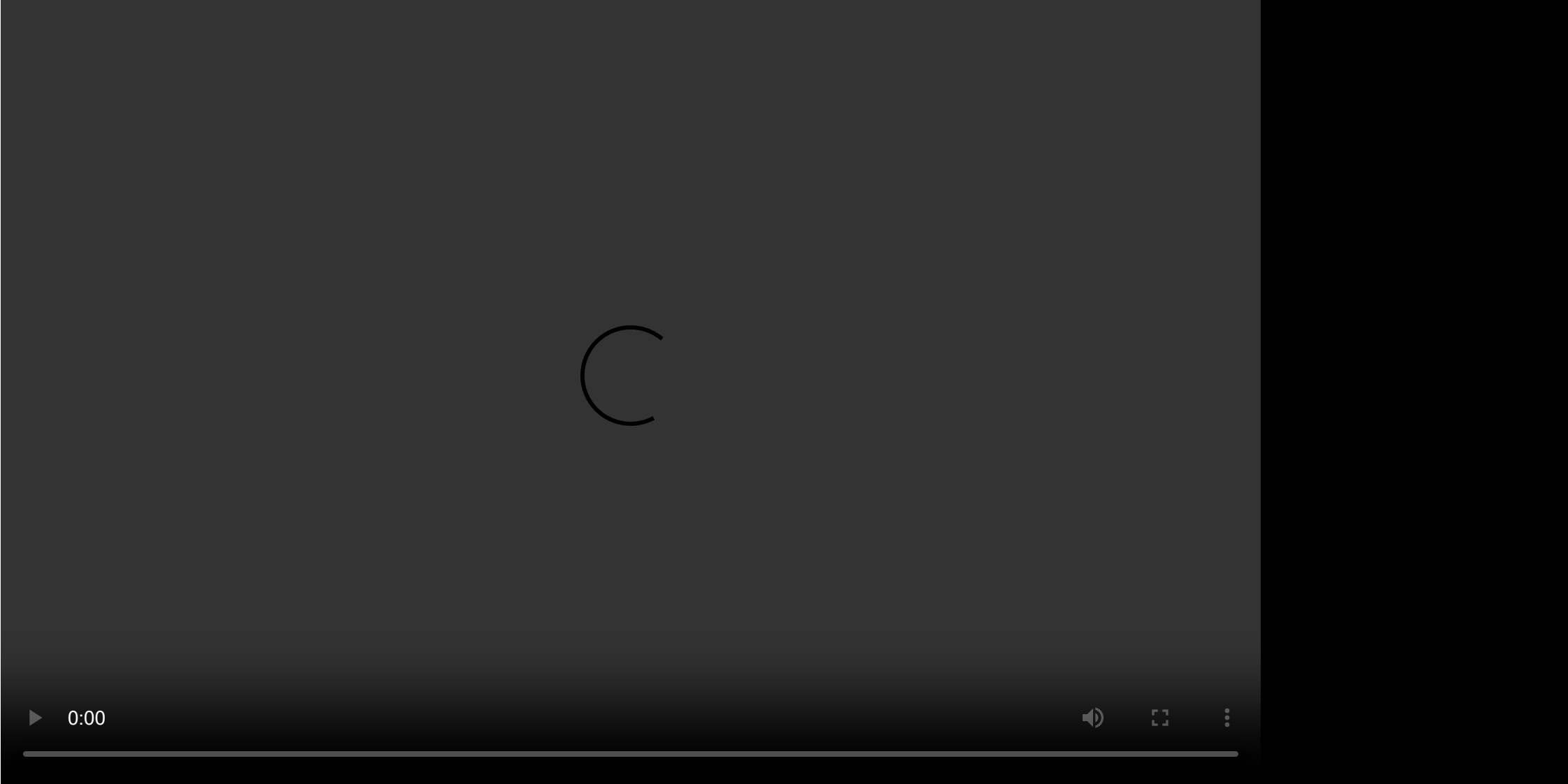
What is Machine Learning?

Machine Learning is a field of study that gives computers the ability to learn without being explicitly programmed. It's a subset of Artificial Intelligence that focuses on building systems that can learn from and make decisions based on data.

ML Overview

Key Characteristics

- Data-driven approach
- Pattern recognition
- Statistical methods
- Iterative learning



Interactive Example

Here's a simple example of how ML works:

```
from sklearn.datasets import make_blobs
import matplotlib.pyplot as plt

---

# Generate sample data
X, y = make_blobs(n_samples=100, centers=2, random_state=42)

---

# Plot the data
plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Sample ML Dataset")
plt.show()
```

Example: Creating a Simple Dataset

You can run the following code in your browser using Pyodide:

```

<div>
  <h3>Interactive Example</h3>
  <button onclick="runExample()">Run Example</button>
  <pre id="output"></pre>
</div>

---

<script>
  async function runExample() {
    let pyodide = await loadPyodide();
    let code = `
import numpy as np
import matplotlib.pyplot as plt

---

# Generate sample data
X = np.random.rand(100, 2)
y = np.array([1 if x[0] + x[1] > 1 else 0 for x in X])

---

# Plot the data
plt.scatter(X[:, 0], X[:, 1], c=y, cmap='viridis')
plt.title("Sample ML Dataset")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.show()
`;
    await pyodide.runPythonAsync(code);
  }
</script>

```